

Assignment-2

Step-1: Load CSV manually and store columns

```
In [10]: import csv

# open the file
f = open('Assignment 2.csv', 'r')
r = csv.reader(f)
next(r)

Out[10]: ['Day', 'Hour', 'consumption(w)']

In [11]: days = []
hours = []
consumptions = []

In [12]: # store each column in its own list
for row in r:
    days.append(int(row[0]))
    hours.append(int(row[1]))
    consumptions.append(float(row[2]))

f.close()
```

Step-2: Group consumption by hour

```
In [13]: hour_dict = {}

for i in range(len(hours)):
    h = hours[i]
    c = consumptions[i]
    if h not in hour_dict:
        hour_dict[h] = []
    hour_dict[h].append(c)

In [14]: # Step 2.1 & 2.2: Compute averages and peaks
hour_avg = {}
hour_peak = {}

In [15]: for h, vals in hour_dict.items():
    hour_avg[h] = sum(vals) / len(vals)
    hour_peak[h] = max(vals)

In [16]: # Step 2.3: Print
print("Hour-wise consumption:")
for h in sorted(hour_dict.keys()):
    print(f"Hour {h}: Avg={hour_avg[h]:.2f}, Peak={hour_peak[h]}")
```

Hour-wise consumption:
Hour 1: Avg=550.00, Peak=800.0
Hour 2: Avg=500.00, Peak=700.0
Hour 3: Avg=537.50, Peak=650.0

Step 3: Group consumption by day

```
In [17]: day_dict = {}
```

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In [18]: for i in range(len(days)):
          d = days[i]
          c = consumptions[i]
          if d not in day_dict:
              day_dict[d] = []
          day_dict[d].append(c)
```

```
In [19]: # Step 3.2 & 3.3: Total and daily average
          day_total = {}
          day_avg = {}
```

```
In [20]: for d, vals in day_dict.items():
          day_total[d] = sum(vals)
          day_avg[d] = sum(vals) / len(vals)
```

Results

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In [21]: print("\nDay-wise consumption:")
          for d in sorted(day_dict.keys()):
              print(f"Day {d}: Total={day_total[d]:.2f}, Avg={day_avg[d]:.2f}")
```

Day-wise consumption:
Day 7: Total=2700.00, Avg=450.00
Day 8: Total=3650.00, Avg=608.33